Ultrasonic Guided Waves for Rail Weld Flaws

Abstract.

We evaluate ultrasonic guided waves for finding internal flaws in the thermite welds of continuous rail. Guided waves detected 95% of flaws that were 2 mm deep or more.

Method.

Two piezoelectric transducers clamped to the rail foot send 50 kHz guided waves along the rail, and a flaw reflects part of each wave back. The arrival time of the echo locates the flaw to within 15 cm.

Data.

We tested 120 welds on a decommissioned line. Of these, 40 contained machined flaws between 1 mm and 6 mm deep, and 80 were intact.

Note.

A fictional sample paper by L. Okafor and S. Iyer (2023), written to test MAIROS. It is not published research: its data and results are invented.

Results.

Guided waves detected 38 of the 40 flawed welds (95%) and raised 3 false alarms on the 80 intact welds. A crew inspected 1 km of track in about 40 minutes. Flaws shallower than 2 mm were missed in 7 of 12 cases.

Limitations.

The main limitation is that the transducers need direct contact with the rail and a gel couplant, which slows inspection. Surface cracks on the rail head were outside the scope of this test.

Open questions.

Future work should test whether guided-wave results can be combined with camera images of the same rail section. It remains unknown how well the method works on curved track.

Conclusion.

Guided waves find deep internal weld flaws reliably, but they are slow to deploy.